A cellular prior enables compact and transferable single-cell inference with CellPFN

Philipp Flotho[1,2], Matthias Flotho[1,2],
Pirya Rani[1,2], Andreas Keller[1,2]

[1]Chair for Clinical Bioinformatics, Saarland University, Saarbrücken, Germany
[2]Helmholtz Institute for Pharmaceutical Research Saarland (HIPS), Helmholtz Centre for Infection Research, Saarbrücken, Germany

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Abstract

Single-cell prediction often requires learning from limited labelled context across heterogeneous experiments. We developed CellPFN, a prior-data fitted network trained on a procedural cellular prior varying biological signals, donor structure and measurement noise. The 12.33-million-parameter model uses task-local gene correspondence without persistent gene identities or target-task gradient updates. After 384,000 scheduled synthetic episodes, CellPFN reached 72.9% mean balanced accuracy across four cell-typing panels and three support budgets, compared with 66.8–70.1% for released TabICLv2 and Kumo large under their tested input configurations. Cellular continued training improved three tabular models' one-shot balanced accuracy by 5.6–10.8 percentage points. Ablations revealed support-dependent benefits and tradeoffs. CellPFN also accepted transcriptome-scale inputs and yielded gene-deletion explanations recovering established cell-type markers. **The framework further identifies neuropathology-associated cellular states and reproducible predictive expression signatures across independent cohorts.**[O04] These results support cellular task priors for learning biological inference with limited labelled data.

Single-cell atlases provide increasingly broad reference maps, yet annotating a new experiment often requires inference from a small labelled set collected under different biological and technical conditions. More profiled cells do not necessarily supply more independent donors or more varied experimental settings. In supervised single-cell annotation, increasing donor diversity can contribute more to generalization than increasing cell count alone[1]. This distinction motivates models that learn how to interpret limited experimental context across heterogeneous tasks.

Single-cell foundation models such as Geneformer, scGPT, scFoundation, UCE and TranscriptFormer learn reusable representations from large collections of measured profiles[2–6]. Their breadth

supports transfer across biological settings, but large-scale pretraining does not guarantee an advantage on each downstream task. Evaluations have found limitations in frozen representations and saturation as pretraining collections expand[7,8]; recent preprint evidence also shows that normalization and linear methods can remain competitive across several benchmarks[9]. These findings motivate a complementary question: can learning improve the inference rule that combines a few labelled examples, even when the expression measurements themselves already provide useful features?

Meta-learning already supports transfer across heterogeneous single-cell experiments, as demonstrated by MARS[10]. Prior-data fitted networks (PFNs) provide a complementary way to learn inference rules. They are trained across prediction problems drawn from a generative prior, amortizing support-conditioned inference into a forward pass[11]. TabPFN and TabICL demonstrate that synthetic task distributions can produce effective general-purpose tabular learners[12–15]. Recent MapPFN and PerturbPFN work further applies prior-based learning to cellular perturbation prediction[16,17]. For transcriptomic classification, the opportunity is to encode the variation encountered across cellular experiments directly into the training prior, while accommodating thousands of sparse features and scarce labels.

Single-cell simulators already model count distributions, dependence, donor variation and regulatory dynamics through complementary statistical and mechanistic approaches[18–25]. Learned generative models also support conditional synthesis and augmentation[26,27]. For PFN training, the design objective extends beyond reproducing a reference dataset to generating diverse labelled inference problems. Controlled variation in signal strength, hidden subpopulations, donor effects and measurement noise can expose a learner to experimental configurations without requiring each configuration to be measured and annotated. Synthetic-to-real learning in optical flow and face analysis provides precedents for learning from known simulated structure[28,29]. Nevertheless, simulator fidelity depends on the statistic and scenario being evaluated[30,31]. The relevant test is whether the induced prior teaches inference that transfers to real biological data.

Here we introduce CellPFN and its procedural cellular prior. CellPFN preserves gene correspondence within each task without persistent gene identities across tasks. We test whether a compact model can achieve competitive biological inference with limited synthetic training exposure, and whether the same simulator can improve released tabular models through continued training. Quantitative ablations examine which architectural and task-distribution choices help, where they fail, and how simulator statistics relate to transfer. We further evaluate transcriptome-scale inputs and query-gene explanations, grounding the prior's utility in inference from limited labelled biological context.

Results

A cellular prior teaches inference from labelled context

We constructed a procedural prior over single-cell prediction problems, varying biological signals together with their observation conditions. The generator samples hierarchical identities, distributed

expression programs, rare markers, correlated variation and donor-dependent states. Donor composition, study and batch shifts, sequencing depth and overdispersed count sampling produce heterogeneous observations. Varying labels and support budgets turns each simulated dataset into multiple inference problems with known labels and controllable nuisance factors. Aggregate statistics from real datasets, including those evaluated here, informed the prior's ranges. All examples used to optimize CellPFN's weights were synthetic.

Each problem comprises labelled support cells and unlabelled queries. Genes retain their correspondence within a task but have no persistent identity across tasks. CellPFN compares each query gene with support-derived class summaries, pools these relations through permutation-invariant statistics[32] and integrates them into direct neural class predictions. A covariance estimate fitted to the support supplies an additional statistical channel when support is sufficient. Prediction requires no gradient updates, and queries do not condition on other query cells.

The reference model has 12.33 million trainable parameters and received 384,000 scheduled episodes from 12,000 simulated dataset banks; 276,828 episodes carried nonzero loss weight. Reused banks supplied distinct task views. The first 320,000 episodes used prior v1 and the remaining 64,000 used v2.2-B. TabICLv2 reports approximately 35 million synthetic pretraining datasets[15], about 90 times CellPFN's scheduled episode count (Supplementary Table S2). These different exposure units contextualize training scale but do not establish a matched-compute advantage.

Compact CellPFN is competitive across biological inference regimes

We evaluated coarse and fine peripheral blood mononuclear cell (PBMC) annotation[33], cortex annotation within the development collection, and cortex queries from held-out ZEBRA studies[34]. Each panel used one, ten or 100 labelled support cells per class with three matched support draws. Comparators included statistical classifiers, released tabular models, and logistic classifiers on Geneformer-V2 and scGPT embeddings[2,3,15,35–37]. PBMC inputs were counts for CellPFN and normalized expression for tabular comparators; cortex used processed expression. Foundation models retained their native input routes, with standardized Geneformer and unscaled scGPT embeddings.

Across the four panels and three support sizes, all-gene CellPFN reached 72.9% mean balanced accuracy, compared with 66.8% for released TabICLv2 and 70.1% for Kumo large using 2k highly variable genes (HVGs; Table 1). Geneformer reached 74.9% through its native input route. These descriptive averages place CellPFN in a competitive range without establishing equivalence or a universal ranking. Supplementary Table S1 reports panel-specific means and three-draw ranges.

Conventional low-rank linear discriminant analysis (LDA) remained strong with abundant labels: at 100 supports per class, its four-panel mean was 84.6%, compared with 85.4% for CellPFN; at ten supports, the means were 75.3% and 79.6%. LDA used processed expression and was not evaluated at one support.

Task-specific differences were larger than the overall differences. For fine PBMC annotation at one support, Geneformer reached 68.8% versus CellPFN's 51.1%; for held-out cortex at ten supports, CellPFN reached 91.9% versus Geneformer's 87.4%. Both foundation models likely

encountered the PBMC cells during unlabelled pretraining. Geneformer-V2 corpus records overlap studies supplying 92.6% of held-out cortex queries, an upper bound on exact-cell overlap. This evidence does not establish label exposure or explain the ranking (Table 1). Model variants were selected on these development panels, with one training seed per arm. **A frozen evaluation on additional, independently held-out cohorts will determine how consistently this competitive performance generalizes.**[O01]

Finer PBMC distinctions remained difficult. At 8,192 HVGs, a separate evaluation gave one-shot balanced accuracies of 62.3% for eight coarse types, 53.9% for 20 fine types and 32.4% across eight harder tasks. The latter separated sibling CD4, CD8, B, NK or broader T-cell subtypes, or all 35 fine-grained labels; their means rose to 63.9% and 80.2% at ten and 100 supports (Supplementary Table S9). Nine CD4 subtypes yielded 23.4% and 79.6% at one and 100 supports. This analysis used ten support draws at one/ten supports and five at 100. The harder tasks were defined post hoc; they are not a prespecified binary difficulty stratum, and the CellPFN export supplies means without draw-level intervals.

The cellular prior also improves pretrained tabular models

The cellular prior also improved already pretrained tabular learners. We continued training released TabICLv2, Kumo small and Kumo large using v2.2-B throughout, with 40,000 cellular tasks and 10,000 generic replay tasks over 2,500 updates per model. At 2k HVGs, one-shot balanced accuracy improved by 7.3, 10.8 and 5.6 percentage points, respectively, averaged over four panels (Table 1). Changes at 100 supports were below one point in magnitude. Benefits were not uniform: fine PBMC accuracy for TabICLv2 at ten supports fell from 73.8% to 64.6%, with all three support draws worsening (Supplementary Table S2).

Cellular continued training of Kumo small improved accuracy by 6.1 points across eight low-support panel–budget combinations, compared with approximately zero change after generic-table continued training. Two further cellular task streams yielded gains of 5.7 and 5.2 points with the same model seed. The control differed in rows and columns per update, so task content was not isolated under matched data volume. Adaptation could also reduce generic-table performance: TabICLv2's one-shot mean balanced accuracy across four generic tables and five split seeds fell from 70.9% to 64.0% (Supplementary Table S2).

Gene-anonymous inference extends to transcriptome-scale inputs

CellPFN accepted more genes than it encountered during training: the widest training tasks contained 8,192 genes, whereas PBMC and cortex evaluations contained 33,538 and 21,158. Shared per-gene computations and symmetric aggregation required no expanded vocabulary. Across the three benchmark draws, balanced accuracy was 71.2% with 2k HVGs, 73.8% with 8,192 and 72.9% with all genes (Table 2). A separate twenty-draw analysis resolved task-specific effects: all genes versus 2k HVGs improved held-out cortex accuracy at ten supports by 7.4 points (95% interval, 5.3–9.6), but reduced one-shot PBMC accuracy by 3.3 points for coarse and 2.0 points for fine labels

(Supplementary Table S3). These paired intervals quantify support-draw variation, not training-run variation.

Released TabICLv2 declined from 66.8% to 48.3% and Kumo large from 70.1% to 66.4% when moving from 2k HVGs to all genes. TabPFN-3.5 refused the all-gene cases at its default feature guard[37,38]. These are outcomes of the tested column-handling configurations, not fundamental architectural limits. Random-gene controls and alternative input views are reported in Supplementary Tables S3 and S9.

Prior and architecture changes have regime-dependent effects

The support-fitted covariance channel had substantial but regime-dependent effects. On fine PBMC annotation at 100 supports, separately trained models with no operator, diagonal variance information or the full operator reached 82.2%, 85.7% and 89.0% balanced accuracy. Full versus no operator gained 6.81 points (95% paired interval, 6.36–7.26); full versus diagonal gained 3.37 points (3.06–3.68), over twenty support draws with one training run per arm (Supplementary Table S4). Nevertheless, the earlier prespecified test failed: the full model lost 1.56 points against the no-operator model on coarse PBMC at ten supports, exceeding the allowed 1.50-point harm threshold. Later descriptive gains do not overturn that outcome.

A ten-candidate prior screen identified learnable-task filtering as the only change meeting its retention rule on the Kumo-small proxy (Supplementary Table S5). After 10,000 cellular tasks, low-support accuracy improved by 1.95 points, with gains of 1.63, 1.62 and 2.61 across three task streams sharing one model seed. No prior-content modification demonstrated benefit. Whole-study gene absence, for example, gained 0.90 points but failed its predictive-log-loss guard, worsening negative log likelihood (NLL) by 0.099 against a 0.03 limit. Other outcomes included neutral and unresolved cases; they do not establish that the corresponding mechanisms are irrelevant.

Learnable-task continuation also improved CellPFN's one-shot accuracy on all four panels: gains were 1.11, 2.42, 2.56 and 3.49 points for coarse PBMC, fine PBMC, development cortex and held-out cortex, respectively (Supplementary Table S6). The held-out cortex gain had a 95% paired interval of 1.73–5.26 over twenty support draws. At 100 supports, NLL worsened on three panels by 0.35–0.63, while improving by 0.05 on fine PBMC. This is a probabilistic-prediction tradeoff, not a calibration-only measurement. Learnable-task continuation processed 1.67 times as many cells per update, precluding attribution to filtering alone.

Simulator diagnostics likewise supported partial agreement with real data (Supplementary Table S7). The median strongest-marker absolute log2 fold change was 6.23 under v2.2-B versus 7.10 in the real reference and 1.94 under v1. Yet eight-class tasks with ten supports were easier: median chance-adjusted nearest-class-mean accuracy was 0.881 versus 0.775, and 37.5% of synthetic banks exceeded the real maximum. These reference statistics informed prior calibration, and v2.2-B supplied only the final 64,000 CellPFN episodes. They therefore contextualize transfer without establishing that increased realism caused better predictions.

Sex prediction varies across evaluation settings

CellPFN also predicted sex in ZEBRA, but performance depended on the evaluation setting. On one frozen oligodendrocyte case, it achieved 100% balanced accuracy over 50 query cells and over ten query GSM samples, using 100 support cells per sex. Support and query GSMs were disjoint; GSM identifiers do not guarantee distinct donors. A later positive control restricted studies to those measuring *XIST* and at least four Y-linked genes. With 8,192 HVGs, mean unit-level AUC was 0.977 at four support units per sex and 1.000 at 16, across ten splits; units included age-by-sex sample groups. The broader sex task yielded 0.507 and 0.538 at those budgets (Supplementary Table S9). Study sets and gene universes also differed, so this contrast does not isolate the effect of sex-linked genes. **Gene-level attribution of these sex predictions will establish which expression features the model uses; the available results do not yet identify those features.**[O06]

Query-gene deletion recovers cell-type marker structure

Gene anonymity still permits gene-level explanations because feature names can be restored after prediction. DeepSets leave-one-gene-out (DS-LOO) approximates the effect of zeroing one query gene on the logit margin between the predicted class and its original runner-up, holding support and fitted covariance fixed. Unlike gradient attribution along an input path[39], the target is a finite deletion. One attribution sweep combines pooled gene statistics with head-gradient evaluations and depth/detection corrections, avoiding a separate full-model forward pass per deleted gene. It is not literally one forward pass, and its target is a model intervention rather than a biological knockout.

Against exact deletions in six cases and two model states, case-level medians of within-cell rank correlation ranged from 0.979 to 0.998; reference-model relative L2 errors were 0.5–4.9%. Validation used selected genes and correctly classified cells in full precision (Supplementary Table S8). **All-gene explanations cost approximately 10–13 single-gene exact forward passes. For the PBMC example, DS-LOO took 67.9 seconds; the corresponding 61-hour exhaustive-deletion runtime was extrapolated from individual forward timings. Shared-GPU timings remain provisional.**[O05]

The explanations recovered recognizable cell-type markers without gene-name embeddings. In PBMC class-margin rankings at 100 supports, B-cell markers *CD79A* and *MS4A1* ranked first and second; *IL7R* ranked second for CD4 T cells, *CD8A* second for CD8 T cells, and *GNLY*/*NKG7* second/seventh for natural killer cells[33,40]. In development cortex, oligodendrocyte-associated *PLP1* ranked eighth for glia, neuronal *SYT1* seventh for neural cells, and endothelial *CLDN5* sixteenth for vascular cells[41]. These are selected lineage-associated matches, not an enrichment test (Supplementary Table S8).

Removing each PBMC query's 50 highest-ranked genes reversed its original runner-up margin in 45.0% of cells, versus 0.3% for random expressed genes. Class-level top-20 overlap across support configurations was 0.43–0.56, indicating context sensitivity. Thus the rankings explain classifier decisions without establishing perturbation responses or universal marker recovery.

A provisional application examines neuropathology-associated expression

In a provisional donor-level application to ROSMAP-derived single-nucleus data[42], microglial predictions distinguished CERAD 4-versus-1 neuropathology groups with an AUC of 0.745 and a reported corrected 95% repeated-split interval of 0.700–0.790. Case and control support was matched within sex (Supplementary Table S9). The result awaits a donor-label audit and frozen rerun; it measures association, not prospective diagnosis or clinical utility.[O02]

Applying query-gene deletion to the corrected disease analysis and testing the findings in independent cohorts will test whether this approach can connect donor-level prediction to reproducible cellular expression signatures.[O03]

Table 1: Biological classification and cellular-prior transfer.
Model comparison: balanced accuracy (%)

Panel, support/class	CellPFN	TabICLv2		Kumo small		Kumo large		TabPFN 3.5	Geneformer V2 (316M)	scGPT	Low-rank LDA
	All genes	Released	+ prior	Released	+ prior	Released	+ prior	Released	+ LR	+ LR	All genes
PBMC coarse, 1	60.1	50.2	62.3	45.8	**62.8**	58.9	62.2	34.3	**67.1***	64.9*	–
PBMC coarse, 10	84.0	81.4	**85.2**	82.2	83.3	81.7	82.5	82.6	**88.5***	77.5*	79.4
PBMC coarse, 100	91.9	92.4	**93.4**	92.6	92.3	92.8	92.8	91.5	92.7*	80.7*	90.0
PBMC fine, 1	**51.1**	30.0	29.1	44.1	49.0	46.3	50.9	33.0	**68.8***	55.7*	–
PBMC fine, 10	79.0	73.8	64.6	80.0	80.5	**83.3**	82.2	79.3	**84.9***	65.9*	70.3
PBMC fine, 100	89.0	83.6	85.3	88.3	88.7	**89.3**	89.2	86.9	**90.4***	72.0*	87.8
ZEBRA development, 1	41.4	39.5	44.6	31.2	41.9	40.1	42.8	38.4	43.1$^{(*)}$	**47.6**	–
ZEBRA development, 10	**63.6**	59.9	60.6	57.5	59.2	56.8	60.6	59.3	61.1$^{(*)}$	60.2	60.6
ZEBRA development, 100	**66.1**	65.3	65.3	65.1	64.9	65.1	65.8	65.5	65.4$^{(*)}$	65.8	66.0
ZEBRA held-study, 1	62.8	49.8	62.7	48.8	59.3	49.1	60.8	54.1	58.4*	**65.5**	–
ZEBRA held-study, 10	**91.9**	83.2	85.8	84.8	86.8	85.1	85.9	85.3	87.4*	87.2	91.0
ZEBRA held-study, 100	**94.7**	92.9	93.2	93.7	92.9	93.1	93.4	93.9	90.3*	89.5	**94.7**
Overall	**72.9**	66.8	69.4	67.8	71.8	70.1	72.4	67.0	**74.9***	69.4*	–

Means over three support draws. Bold identifies the highest row mean overall and among unflagged entries, including ties at displayed precision; gray marks starred exposure flags. Rankings are descriptive, not significance claims. Overall equally weights twelve panel–budget combinations; dashes denote unevaluated settings, and LDA has no overall mean. ZEBRA development and held-study query panels are distinct. Tabular models use 2k HVGs; CellPFN and LDA use all genes. CellPFN uses PBMC counts and processed ZEBRA values; tabular models and LDA use processed values. Geneformer/scGPT use native embeddings with logistic regression (LR); scGPT uses the declared unstandardized route. "+ prior" denotes released weights adapted with 40,000 cellular tasks plus generic replay. Panel-specific draw ranges appear in Table S1. HVG, highly variable gene.
* flags likely pretraining exposure: PBMC cells are assessed as likely present for Geneformer-V2 (316M)[35] and scGPT. Geneformer-V2 corpus records overlap source studies supplying 92.6% of ZEBRA held-study query cells; this is an upper bound on exact-cell overlap, not verified cell matching. $^{(*)}$ denotes inferred possible overlap of control cells in ZEBRA development (at most 22.3%). These marks concern cell-expression exposure, not evidence that target labels were seen. Overall starred means include flagged evaluations. Unmarked scores do not certify absence of exposure outside the audited Geneformer/scGPT corpora.

Table 2: CellPFN gene-panel ablation.
Balanced accuracy (%)

Panel	1 support/class			10 supports/class			100 supports/class		
	2k HVGs	8,192 HVGs	All genes	2k HVGs	8,192 HVGs	All genes	2k HVGs	8,192 HVGs	All genes
PBMC coarse	60.4	**61.1**	60.1	82.6	82.8	**84.0**	90.6	**92.0**	91.9
PBMC fine	53.0	**54.3**	51.1	79.8	**81.3**	79.0	87.4	**90.3**	89.0
ZEBRA development	43.2	**45.8**	41.4	60.8	63.4	**63.6**	64.2	65.1	**66.1**
ZEBRA held-study	61.1	**65.0**	62.8	80.9	**92.0**	91.9	89.8	92.7	**94.7**

Means over three support draws for the same CellPFN checkpoint and query panels used in Table 1. Bold identifies the highest mean within each panel and support-budget group, including ties at displayed precision. Rankings are descriptive, not significance claims. CellPFN uses PBMC counts and processed ZEBRA values; ZEBRA development and held-study query panels are distinct. HVG, highly variable gene. Table S3 reports the separate twenty-draw analysis, paired intervals and random-gene controls.

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Supporting quantitative results

These tables accompany the main Results and retain negative findings, exploratory analyses and limits on replication. Unless stated otherwise, intervals describe variation over support draws for fixed models and query panels; they do not represent independent model training or population-level biological replication.

Table S1: Panel-specific benchmark performance and contextual baselines.

Model	1 support/class	10 supports/class	100 supports/class
PBMC coarse			
CellPFN	60.1 [52.5, 70.9]	84.0 [81.9, 85.9]	91.9 [91.6, 92.2]
TabICLv2	50.2 [41.3, 58.4]	81.4 [79.4, 83.1]	92.4 [91.3, 93.4]
TabICLv2 + cellular prior	62.3 [48.1, 72.5]	**85.2** [81.9, 88.1]	**93.4** [93.1, 94.1]
Kumo small	45.8 [40.3, 50.3]	82.2 [80.6, 83.4]	92.6 [91.6, 94.7]
Kumo small + cellular prior	**62.8** [52.8, 69.7]	83.3 [80.9, 86.3]	92.3 [91.6, 93.1]
Kumo large	58.9 [46.3, 65.6]	81.7 [78.8, 83.8]	92.8 [91.9, 93.8]
Kumo large + cellular prior	62.2 [50.3, 70.0]	82.5 [79.4, 85.0]	92.8 [91.9, 93.4]
TabPFN-3.5	34.3 [30.9, 38.4]	82.6 [80.6, 83.8]	91.5 [90.0, 92.5]
Geneformer-V2 + LR	**67.1*** [58.1, 74.7]	**88.5*** [86.9, 90.3]	92.7* [91.6, 93.4]
scGPT + LR	64.9* [53.1, 72.2]	77.5* [75.6, 78.8]	80.7* [80.3, 81.3]
Low-rank LDA	–	79.4 [78.8, 80.0]	90.0 [89.7, 90.3]
CellTypist	39.8 [33.8, 47.5]	77.5 [76.9, 78.1]	87.8 [87.5, 88.1]
SingleR	56.7 [52.8, 60.9]	78.2 [74.1, 80.9]	85.9 [85.0, 86.6]
scANVI inductive	20.0 [11.3, 32.5]	40.7 [26.6, 51.3]	86.7 [86.6, 86.9]
scANVI transductive	40.5 [36.3, 44.1]	69.1 [65.9, 74.7]	89.1 [88.1, 90.9]
Logistic regression	37.5 [34.4, 43.4]	80.5 [79.1, 81.6]	91.7 [90.6, 93.1]
PBMC fine			
CellPFN	**51.1** [44.8, 55.4]	79.0 [77.6, 80.4]	89.0 [88.6, 89.1]
TabICLv2	30.0 [26.8, 34.3]	73.8 [72.1, 75.6]	83.6 [82.9, 84.5]
TabICLv2 + cellular prior	29.1 [26.4, 32.3]	64.6 [61.0, 67.9]	85.3 [84.3, 86.1]
Kumo small	44.1 [43.6, 44.6]	80.0 [79.4, 80.4]	88.3 [88.1, 88.4]
Kumo small + cellular prior	49.0 [45.9, 52.3]	80.5 [80.0, 81.0]	88.7 [87.6, 89.6]
Kumo large	46.3 [42.3, 50.0]	**83.3** [82.8, 83.6]	**89.3** [89.1, 89.5]
Kumo large + cellular prior	50.9 [47.0, 55.8]	82.2 [81.4, 83.5]	89.2 [89.1, 89.4]
TabPFN-3.5	33.0 [29.4, 36.5]	79.3 [77.4, 80.4]	86.9 [86.5, 87.3]
Geneformer-V2 + LR	**68.8*** [66.6, 70.9]	**84.9*** [84.3, 85.5]	**90.4*** [90.0, 90.9]
scGPT + LR	55.7* [51.3, 58.1]	65.9* [65.3, 67.1]	72.0* [71.5, 72.9]
Low-rank LDA	–	70.3 [70.0, 70.5]	87.8 [87.5, 87.9]
CellTypist	33.9 [26.8, 37.8]	69.7 [68.6, 70.8]	84.2 [83.9, 84.6]
SingleR	46.0 [42.1, 51.8]	68.6 [67.0, 70.1]	83.3 [82.6, 83.9]
scANVI inductive	21.7 [14.4, 26.1]	51.8 [50.8, 53.4]	83.9 [81.6, 85.3]
scANVI transductive	40.8 [39.5, 41.5]	70.8 [68.8, 74.0]	86.2 [85.8, 86.5]
Logistic regression	39.0 [34.0, 46.8]	73.4 [71.8, 75.9]	85.1 [84.4, 85.6]
ZEBRA development			
CellPFN	41.4 [33.1, 47.4]	**63.6** [61.9, 65.6]	**66.1** [65.8, 66.4]
TabICLv2	39.5 [33.4, 42.6]	59.9 [57.9, 62.2]	65.3 [64.9, 65.7]
TabICLv2 + cellular prior	44.6 [35.9, 57.1]	60.6 [57.4, 62.6]	65.3 [65.1, 65.5]
Kumo small	31.2 [26.6, 36.4]	57.5 [50.5, 61.7]	65.1 [64.5, 65.7]
Kumo small + cellular prior	41.9 [34.9, 51.2]	59.2 [53.4, 62.3]	64.9 [63.8, 65.6]
Kumo large	40.1 [38.2, 41.5]	56.8 [49.2, 61.2]	65.1 [64.9, 65.3]
Kumo large + cellular prior	42.8 [37.3, 52.3]	60.6 [55.6, 63.4]	65.8 [65.6, 66.1]
TabPFN-3.5	38.4 [32.8, 43.3]	59.3 [56.6, 60.8]	65.5 [64.8, 66.0]

Table S1 (continued)

Model	1 support/class	10 supports/class	100 supports/class
Geneformer-V2 + LR	43.1$^{(*)}$ [34.8, 50.2]	61.1$^{(*)}$ [55.3, 66.7]	65.4$^{(*)}$ [65.3, 65.5]
scGPT + LR	**47.6** [43.7, 55.0]	60.2 [53.2, 66.5]	65.8 [65.6, 66.1]
Low-rank LDA	–	60.6 [58.4, 62.0]	66.0 [65.8, 66.4]
CellTypist	34.9 [32.6, 37.3]	57.8 [48.6, 64.0]	63.9 [63.5, 64.6]
SingleR	44.4 [34.4, 53.3]	61.7 [59.8, 63.5]	63.9 [63.2, 64.2]
scANVI inductive	–	–	–
scANVI transductive	–	–	–
Logistic regression	34.7 [28.1, 39.6]	57.4 [48.7, 62.4]	65.5 [65.1, 65.8]
ZEBRA held-study			
CellPFN	62.8 [45.2, 77.1]	**91.9** [88.6, 94.7]	**94.7** [94.3, 95.0]
TabICLv2	49.8 [46.6, 55.2]	83.2 [75.5, 88.1]	92.9 [92.1, 93.5]
TabICLv2 + cellular prior	62.7 [37.8, 81.0]	85.8 [81.9, 88.7]	93.2 [92.2, 94.0]
Kumo small	48.8 [31.7, 65.9]	84.8 [81.3, 86.7]	93.7 [92.3, 94.7]
Kumo small + cellular prior	59.3 [41.8, 73.5]	86.8 [83.4, 88.7]	92.9 [91.1, 94.4]
Kumo large	49.1 [39.6, 56.6]	85.1 [80.3, 88.8]	93.1 [91.7, 94.5]
Kumo large + cellular prior	60.8 [39.1, 75.8]	85.9 [81.4, 88.6]	93.4 [93.2, 93.6]
TabPFN-3.5	54.1 [48.0, 65.3]	85.3 [84.9, 85.8]	93.9 [92.5, 95.3]
Geneformer-V2 + LR	58.4* [44.2, 72.5]	87.4* [83.8, 91.7]	90.3* [88.4, 92.6]
scGPT + LR	**65.5** [52.5, 80.7]	87.2 [80.6, 93.1]	89.5 [88.9, 90.2]
Low-rank LDA	–	91.0 [89.6, 93.8]	**94.7** [94.0, 95.4]
CellTypist	44.4 [41.6, 47.5]	87.8 [80.4, 92.5]	91.4 [90.6, 92.3]
SingleR	56.8 [38.4, 71.1]	86.6 [79.7, 90.8]	89.1 [86.8, 90.6]
scANVI inductive	–	–	–
scANVI transductive	–	–	–
Logistic regression	45.0 [42.1, 48.1]	73.5 [70.7, 76.0]	90.6 [90.0, 91.5]

Values are mean [minimum, maximum] balanced accuracy (%) over the same three support draws, not confidence intervals. Within each panel and support budget, bold identifies the highest mean overall and among unflagged comparators, including ties at displayed precision. Gray means and ranges carry the exposure stars defined in Table 1; unflagged entries do not certify absence of exposure. Inputs follow Table 1; CellTypist, SingleR and low-rank LDA use all genes, whereas logistic regression and scANVI use 2k HVGs. CellTypist[43] is fitted to the support cells; SingleR[44] uses the declared reference route. The scANVI[45] transductive route additionally uses the unlabelled query pool and is contextual rather than an inductive comparison. Dashes are missing evaluations, never zero accuracy. All selected fitted model checkpoints have one training seed; support-draw variation does not measure training uncertainty. Geneformer-V2 uses the 316M checkpoint. Source-study corpus overlap does not establish exact-cell or target-label exposure. Native gene vocabularies and training histories differ. Family variants were selected in the export and the evaluation is not a newly frozen independent benchmark.

Table S2: Training exposure, prior transfer and generic-task retention.

Model	Quantity	Count
CellPFN	Parameters	12329889
CellPFN	Optimizer updates	96000
CellPFN	Simulated banks	12000
CellPFN	Scheduled episodes	384000
CellPFN	Loss-bearing episodes	276828
TabICLv2	Parameters	27552258
TabICLv2	Additional cellular tasks	40000
TabICLv2	Adaptation updates	2500
Kumo small	Parameters	27458266
Kumo small	Additional cellular tasks	40000
Kumo small	Adaptation updates	2500
Kumo large	Parameters	213668250
Kumo large	Additional cellular tasks	40000
Kumo large	Adaptation updates	2500

CellPFN banks are reused for multiple episodes; scheduled and loss-bearing episodes are distinct. Adaptation uses 40,000 cellular tasks and 10,000 generic replay tasks per arm. Released-model pretraining precedes this adaptation and is not included in these counts. TabICLv2 reports 35 million synthetic pretraining datasets[15]; this published count is distinct from CellPFN banks and scheduled episodes. These units do not establish a matched-compute comparison. **Adapted minus released balanced accuracy (percentage points).**

Model	1 support	10 supports	100 supports
TabICLv2	+7.30	-0.50	+0.77
Kumo small	+10.79	+1.31	-0.22
Kumo large	+5.58	+1.08	+0.26

Kumo-small low-support change relative to released weights.

Training stream	Mean change (percentage points)
Cellular task stream 1	+6.05
Cellular task stream 2	+5.73
Cellular task stream 3	+5.17
Generic-table control	-0.05

The low-support average weights eight panel–budget combinations equally (three support draws each). The three cellular runs vary the task stream with the same model seed. The generic control changes rows and columns per update, so this is not a matched-data or matched-cost contrast. No corresponding generic-only control was run for TabICLv2 or Kumo large.

Retention on four generic classification datasets.

Model	One-shot BA	One-shot accuracy	All-budget BA	All-budget accuracy
TabICLv2 released	70.9	69.3	87.8	87.2
TabICLv2 adapted	64.0	62.0	84.8	84.3

Balanced accuracy (BA) and ordinary accuracy (%) equally weight breast cancer, digits, iris and wine, with five split seeds each. The all-budget mean additionally weights 1, 10 and the full training split equally. These exploratory retention checks do not estimate universal tabular performance.

Table S3: Gene-panel scaling and random-gene controls.

Panel	Support/class	All minus 2k HVGs (pp; 95% CI)
PBMC coarse	1	-3.31 [-5.20, -1.42]
PBMC coarse	10	+0.91 [-0.11, 1.92]
PBMC coarse	100	+1.31 [0.92, 1.71]
PBMC fine	1	-1.96 [-3.27, -0.65]
PBMC fine	10	-0.55 [-1.25, 0.15]
PBMC fine	100	+1.81 [1.44, 2.18]
ZEBRA development	1	+0.26 [-2.58, 3.10]
ZEBRA development	10	+2.52 [1.66, 3.39]
ZEBRA development	100	+1.30 [0.98, 1.61]
ZEBRA held-study	1	+1.11 [-2.05, 4.27]
ZEBRA held-study	10	+7.41 [5.26, 9.57]
ZEBRA held-study	100	+4.38 [3.69, 5.07]

Paired differences in balanced accuracy (percentage points) use twenty support draws per panel–budget combination. Intervals are the exported paired 95% t intervals over support draws. They condition on the evaluated checkpoint and query set; they do not cover training, donor-population or panel-selection uncertainty. **Mean balanced accuracy across the twelve panel–budget combinations.**

Gene panel	Mean balanced accuracy (%)
Random 2,000 genes	62.9
2k HVGs	72.4
Random 8,192 genes	69.4
8,192 HVGs	**74.1**
All genes	73.7

Bold identifies the highest descriptive gene-panel average, including ties at displayed precision; paired contrast intervals above are not ranked. These descriptive twenty-draw averages differ from the three-draw headline benchmark in Table 1. Random-panel controls assess the delivered panels and cannot isolate every effect of feature selection or gene coverage.

Table S4: Operator ablation and the prespecified decision.

Panel	Support/class	Full minus no operator (pp; 95% CI)
PBMC coarse	1	+0.20 [-0.23, 0.64]
PBMC coarse	10	-0.50 [-1.24, 0.24]
PBMC coarse	100	+2.69 [2.35, 3.02]
PBMC fine	1	+0.61 [0.19, 1.03]
PBMC fine	10	+4.18 [3.48, 4.87]
PBMC fine	100	+6.81 [6.36, 7.26]
ZEBRA development	1	-0.29 [-0.60, 0.01]
ZEBRA development	10	+1.97 [0.99, 2.95]
ZEBRA development	100	+2.07 [1.77, 2.37]
ZEBRA held-study	1	-1.35 [-1.98, -0.73]
ZEBRA held-study	10	+2.71 [1.39, 4.04]
ZEBRA held-study	100	+2.33 [1.86, 2.80]

Twenty paired support draws, one training seed per arm, all genes. Intervals are exported paired 95% t intervals over support draws. Full and no-operator checkpoints share the parent, prior and task stream. These later descriptive evaluations do not replace the prespecified decision below.

Operator structure across support budgets.

Panel	Support/ class	None	Diagonal	Full	Full minus diagonal (pp; 95% CI)
PBMC coarse	1	59.81	59.09	60.02	+0.92 [0.39, 1.45]
PBMC coarse	10	85.84	85.69	85.34	-0.34 [-0.86, 0.18]
PBMC coarse	100	89.14	88.89	91.83	+2.94 [2.64, 3.24]
PBMC fine	1	51.30	51.38	51.91	+0.53 [0.13, 0.93]
PBMC fine	10	75.64	78.48	79.81	+1.33 [0.87, 1.79]
PBMC fine	100	82.22	85.66	89.03	+3.37 [3.06, 3.68]
ZEBRA development	1	46.62	46.45	46.33	-0.12 [-0.49, 0.24]
ZEBRA development	10	61.36	63.20	63.33	+0.13 [-0.48, 0.75]
ZEBRA development	100	63.70	64.74	65.77	+1.03 [0.89, 1.18]
ZEBRA held-study	1	66.53	65.84	65.17	-0.67 [-1.08, -0.25]
ZEBRA held-study	10	88.59	91.11	91.31	+0.19 [-0.32, 0.70]
ZEBRA held-study	100	92.10	93.57	94.43	+0.86 [0.40, 1.32]

Values are mean balanced accuracy (%). The diagonal ablation is a trained per-gene-variance-only operator arm; it shares the parent, prior and task stream with the full-operator arm. The printed matrices use all genes. The source-value index retains additional HVG and random-panel analyses.

Prespecified three-draw readout.

Panel	Support/class	Full minus none (pp)	Harm threshold (pp)
PBMC coarse	10	-1.56	1.50
PBMC coarse	100	+3.02	1.50
ZEBRA held-study	10	+3.30	2.72

The overall test was not supported because coarse-PBMC ten-support accuracy crossed the prespecified harm threshold (N3). Turning the full operator off at inference isolated a harmful -0.94-point operator contribution in that setting. Its donor-enumeration interval has approximately 80% coverage, not 95%, and is not used as a 95% interval here. Three frozen support draws and four query donors informed the PBMC test; the later twenty-draw estimates describe support sensitivity rather than overturning the decision.

Table S5: Complete ten-candidate prior screen.

Arm/base	Change	r42	r43	r44	Mean	Decision
LR / b0	Learnable-only tasks	+1.63	+1.62	+2.61	+1.95	Keep
A5 / lr	Fewer/weaker markers and closer sibling classes	-1.52	-1.05	-1.54	-1.37	Drop: guard
A6 / b0	Remove signed-covariance task lane	-0.19	+0.74	+1.16	+0.57	Neutral
A7 / b0	Donor dominance	+0.16	-0.21	+0.65	+0.20	Unresolved
A1 / b0	More donor-rich tasks	-0.0008	-0.0080	-0.0133	-0.0074	Routing only
A3 / a1	Sex mechanisms	-0.0089	-0.0089	+0.0357	+0.0060	Routing only
A4 / lr	Weak metadata tasks	+0.0110	-0.0347	-0.0325	-0.0188	Routing only
A2 / a1age	Adult-age prior	+0.0083	+0.0000	+0.0000	+0.0028	Routing only
A8a / b0	Study-wide gene absence	+0.52	+0.54	+1.64	+0.90	Drop: guard
A8d / b0	Study-specific detection	-0.43	+0.60	+0.87	+0.35	Not testable

All changes are improvement-signed relative to the indicated base. LR, A5, A6, A7, A8a and A8d use balanced-accuracy percentage points; A1, A3 and A2 use donor AUC differences on a 0–1 scale; A4 uses negative log-loss differences (base minus arm, nats). A7 uses the held-study panel; A8a/A8d use low-support cortex; LR/A5/A6 use all low-support core panels. A1/A3/A2 respectively target donor, worse-sex and adult-age AUC. The Kumo-small proxy uses 10,000 cellular tasks, 625 updates and 2k HVGs. The three columns are task-stream replicates from one model seed, not independent model-seed replications. A5 and A4 are compared with LR; A3 with A1; A2 with the matched age-task stream. Other candidates use the pipeline control b0.

LR was the only retained change: its keep threshold was 1.63 points and effective noise band 1.09 points. A5 failed the low-support guard. A7 remained unresolved; omitting the prescribed two extension replicates departed from the frozen rule, although the required keep count was no longer attainable. A1–A4 routing labels do not establish successful donor-level or probability-prediction improvements: those endpoints were outside the proxy's trusted decision scope. A8a failed the log-loss guard (0.099-nat deterioration against a 0.03-nat limit). A8d was not testable because measured headroom (1.257 points) was below the 1.5-point keep threshold; its two extension replicates were omitted under the documented shortcut because the required keep count was no longer attainable. A8d replicate r43 skipped one invalid task, an authorized deviation made after the other two replicate readouts. None of these routing, unresolved or headroom-limited outcomes is evidence that its biological mechanism does or does not work.

Stored-zero mechanism readout (percentage points).

Arm	Mean gap shrinkage	Noise band	Proxy headroom	Keep threshold
A8a	0.072	0.620	1.257	1.500
A8d	0.071	0.620	1.257	1.500

Gap shrinkage is the reduction in the accuracy difference between removing unmeasured genes and retaining stored zeros. The gap did not shrink beyond the noise band; the reference row repeats the three-replicate aggregate, not a single-replicate estimate.

Table S6: Learnable-only versus standard CellPFN continuation.

Panel	Support/class	Accuracy change (pp; 95% CI)	Log-loss change (nats; 95% CI)
PBMC coarse	1	+1.11 [0.45, 1.77]	-0.332 [-0.400, -0.265]
PBMC coarse	10	-1.47 [-2.12, -0.81]	+0.416 [0.337, 0.496]
PBMC coarse	100	-1.56 [-1.94, -1.19]	+0.374 [0.357, 0.391]
PBMC fine	1	+2.42 [1.61, 3.23]	-0.742 [-0.773, -0.710]
PBMC fine	10	+1.06 [0.26, 1.86]	-0.159 [-0.172, -0.145]
PBMC fine	100	+0.50 [0.22, 0.78]	-0.047 [-0.051, -0.042]
ZEBRA development	1	+2.56 [1.11, 4.02]	-0.054 [-0.205, 0.097]
ZEBRA development	10	+1.34 [0.35, 2.33]	+0.281 [0.198, 0.363]
ZEBRA development	100	+0.33 [0.19, 0.48]	+0.629 [0.606, 0.652]
ZEBRA held-study	1	+3.49 [1.73, 5.26]	-0.251 [-0.363, -0.139]
ZEBRA held-study	10	+1.38 [0.72, 2.05]	+0.249 [0.186, 0.312]
ZEBRA held-study	100	+0.24 [-0.02, 0.49]	+0.346 [0.304, 0.388]

Learnable-only (GL1) minus standard (GS1) at 100,000 updates; twenty paired support draws; all genes. Positive accuracy change is favourable, whereas positive log-loss change is unfavourable. Intervals are exported paired 95% t intervals over support draws. Both arms continue the same checkpoint for 4,000 updates with one model seed. Log loss measures probabilistic prediction and is not exclusively a calibration measure.

Realized training exposure during continuation.

Task stream	Mean cells per update
Learnable-only	2667.8
Standard	1596.6

The learnable-only continuation processed approximately 1.67 times as many cells per update. Its gains therefore cannot be attributed to task filtering alone under matched cell exposure. Higher-support log loss worsened on three of four panels but improved on fine PBMC.

Table S7: Simulator realism diagnostics against public biological references.

Statistic	Prior	Synthetic median [5th, 95th percentile]	Real median [minimum, maximum]
Strongest marker (log2 FC)	v1	1.936 [1.408, 3.497]	7.100 [3.310, 9.762]
Strongest marker (log2 FC)	v2.2-B	6.232 [4.273, 10.225]	7.100 [3.310, 9.762]
Class separation (chance-adjusted BA)	v1	0.022 [0.001, 0.120]	0.775 [0.525, 0.916]
Class separation (chance-adjusted BA)	v2.2-B	0.881 [0.522, 0.994]	0.775 [0.525, 0.916]
Within-study donor SD (log2)	v1	0.331 [0.213, 0.489]	0.290 [0.197, 0.391]
Within-study donor SD (log2)	v2.2-B	0.404 [0.291, 0.528]	0.290 [0.197, 0.391]

Synthetic quantile ranges and observed real ranges are descriptive distributions, not confidence intervals. The reference contains public PBMC, cortex and kidney datasets; ROSMAP is excluded. Class separation uses nearest-class-mean balanced accuracy at ten supports per class with eight classes in both synthetic and primary real references.

Sampling units and synthetic tail frequencies by prior.

Statistic	Prior	Synthetic banks	Synthetic units	Real units	Synthetic above real max (%)
Strongest marker (log2 FC)	v1	72	2016	142	0.0
Strongest marker (log2 FC)	v2.2-B	96	2688	142	8.3
Class separation (chance-adjusted BA)	v1	72	72	8	0.0
Class separation (chance-adjusted BA)	v2.2-B	96	96	8	37.5
Within-study donor SD (log2)	v1	96	768	11	37.9
Within-study donor SD (log2)	v2.2-B	128	1024	11	58.9

Units differ by statistic: class-pair comparisons for marker effects, dataset-level summaries for separation and donor-effect units for within-study SD. Units from one simulated bank are not independent biological replicates. The real reference helped calibrate the prior, so this is a reference-comparison diagnostic rather than independent validation. Prior v2.2-B supplied only the final 16.7% of scheduled CellPFN training episodes. Improved moment agreement is not itself evidence of a causal performance benefit.

Table S8: DS-LOO fidelity and support sensitivity.

Case	Full operator: Spearman	No operator: Spearman	Full operator: relative error (%)
PBMC, 10 supports, draw 0	0.997	0.997	2.91
PBMC, 10 supports, draw 1	0.998	0.997	2.31
PBMC, 100 supports	0.998	0.995	1.56
Cortex development, 10 supports	0.993	0.988	0.48
Kidney, counts	0.986	0.979	4.88
Kidney, processed	0.988	0.989	1.17

Corrected-margin DS-LOO versus exact query-only single-gene knockout, in fp32, on the selected evaluated genes and correctly classified cells. Spearman values summarize within-cell rank correlations, aggregated within classes and then across classes. Relative L_2 errors are computed per class and summarized by the median across classes. These are six case summaries per model, not minima across every cell. No one-shot or held-study cortex attribution validation is included.

Comparison	Range of class-mean top-20 overlap
Different PBMC support choices	0.431–0.563
Full versus no operator	0.663–0.878

Overlap is the fraction of common genes among the twenty highest-ranked genes, averaged over classes for each comparison. The support comparisons include both a changed support draw at ten cells per class and a changed support budget; they are not a pure seed-only sensitivity analysis. Agreement with model knockout measures explanation fidelity, not causal biological validation.

Three highest-ranked genes for each explained cell class.

Panel	Cell class	Rank 1	Rank 2	Rank 3
PBMC	B	CD79A	MS4A1	BANK1
PBMC	CD4 T	MAL	IL7R	TRAC
PBMC	CD8 T	CD8B	CD8A	TRBC2
PBMC	DC	FCER1A	CLEC10A	HLA-DQA1
PBMC	Mono	SERPINA1	FTL	CD14
PBMC	NK	KLRF1	GNLY	IL2RB
PBMC	other	GNG11	PF4	PPBP
PBMC	other T	GZMK	TRGC2	KLRB1
ZEBRA	Glia	NPAS3	PCDH9	PPP2R2B
ZEBRA	Immune cell	MBNL1	FMN1	DPYD
ZEBRA	Neural	FGF12	RGS7	RBFOX1
ZEBRA	Vascular cell	PTPRG	MALAT1	LAMA2

Genes are selected directly by their class-mean corrected DS-LOO margin rank, independently of the curated marker checks below. PBMC uses 100 supports per class, draw 0; ZEBRA development uses ten supports per class, draw 0. These are model-ranked features conditional on the support set and class-discrimination target; they are not all canonical lineage markers and do not establish causal effects.

Known-marker ranks in the PBMC reference case.

Cell class	Marker	Corrected DS-LOO rank
B	CD79A	1
B	MS4A1	2
CD4 T	IL7R	2
CD8 T	CD8A	2
DC	FCER1A	1
DC	CLEC10A	2
Mono	CD14	3
NK	GNLY	2
NK	NKG7	7
other	PPBP	3
other T	CD3E	5
DC	LILRA4	4392
Mono	LYZ	33354

The reference uses 100 supports per class and support draw 0, explaining 191 correctly classified query cells across eight broad classes and 33,538 genes. Ranks refer to class-mean corrected margin attribution; smaller is higher. Familiar lineage markers[40] rank highly in several classes, but LYZ in monocytes and LILRA4 in pooled dendritic cells illustrate that marker identity does not guarantee a high rank for this class-discrimination target. These are selected descriptive marker checks, not an enrichment test or causal validation. Gene identities were assigned after inference.

Contextual marker ranks in ZEBRA cell-type classification.

Cell class	Marker	Corrected DS-LOO rank
Glia	PLP1	8
Neural	SYT1	7
Vascular cell	FLT1	5
Vascular cell	CLDN5	16
Immune cell	PTPRC	20710

ZEBRA development, ten supports per class, draw 0. PLP1, SYT1 and CLDN5 provide contextual cell-identity checks[41]. These are cell-type attribution ranks; they are not explanations of the sex-classification experiments in Table S9.

Table S9: Exploratory sensitivity analyses and provisional application.

Panel	Support/class	Processed cortex BA (%)	Reconstructed-count BA (%)
ZEBRA development	1	41.4	51.4
ZEBRA development	10	63.6	63.4
ZEBRA development	100	66.1	64.9
ZEBRA held-study	1	62.8	65.9
ZEBRA held-study	10	91.9	92.9
ZEBRA held-study	100	94.7	93.8

Mean CellPFN balanced accuracy over three support draws. Processed and reconstructed-count cortex have different gene sets (21,158 versus 12,325), and some support identities change. These comparisons therefore do not isolate normalization, count reconstruction or stored-zero effects; the processed representation remains the primary panel.

Removing unmeasured genes while retaining processed values.

Panel	Support/class	Accuracy change (pp; 95% CI)
ZEBRA development	1	+3.62 [0.42, 6.83]
ZEBRA development	10	+0.64 [0.38, 0.90]
ZEBRA development	100	+0.14 [-0.10, 0.38]
ZEBRA held-study	1	+1.04 [-0.67, 2.75]
ZEBRA held-study	10	+0.51 [0.03, 0.99]
ZEBRA held-study	100	-0.47 [-1.66, 0.71]

These Stage-0 contrasts remove genes not measured in every study without changing the retained expression values or re-normalizing their denominator. Values and paired 95% t intervals are the exported summaries over ten support draws. This is a separate diagnostic from the three-draw count-reconstruction comparison above.

Post hoc classifier sensitivity. Standardizing scGPT embeddings before logistic regression changed the PBMC mean from 69.5% to 73.5%. This diagnostic is not substituted for the declared baseline in Table 1.

Post hoc hard-tier boundary. Learnable-only proxy training changed held-study cortex neuronal-subtype accuracy at ten supports by -5.83 percentage points across three task streams (-6.62 to -4.56). Hard-tier tasks were defined after the initial screen and do not change its frozen labels. A8d's hard-tier readout contains only two task-stream replicates.

Early task-mixture gene-view sensitivity at update 98,304.

Gene view	Support draws	PBMC low-support change (pp)
All genes	3	+0.344
8,192 HVGs	10	-0.892

Changes are GL1 minus GS1, equally weighting coarse and fine PBMC at one and ten supports per class. Different draw counts and the early checkpoint prevent treating this as a final, isolated explanation of the task-mixture tradeoff.

Provisional ROSMAP application: sex-matched microglial CERAD discrimination gave donor AUC 0.745 [0.700, 0.790]. The supplied interval summarizes ten repeated splits with the export's correction; it is not independent cohort replication. Label corrections remain pending, so this is not a final or confirmatory biological result.

PBMC annotation levels and fine-grained subtype tasks.

Task	Classes	1 support/class	10 supports/class	100 supports/class
Broad PBMC types (L1)	8	62.3	84.2	91.9
Fine PBMC types (L2)	20	53.9	81.8	90.5
CD4 T siblings (L2)	4	36.3	69.2	85.6
T-cell siblings (L2)	9	27.4	64.4	83.8
CD4 T siblings (L3)	9	23.4	56.3	79.6
CD8 T siblings (L3)	7	39.5	69.4	84.4
B-cell siblings (L3)	6	31.6	63.6	79.8
NK-cell siblings (L3)	6	36.2	64.3	73.0
T-cell siblings (L3)	17	25.0	55.7	76.8
All finest PBMC types (L3)	35	39.8	68.4	78.4
Mean of eight subtype tasks	–	32.4	63.9	80.2

Values are mean balanced accuracy (%) for the reference CellPFN checkpoint with 8,192 HVGs. Broad/fine context rows use the E-fast ten-, ten- and five-draw readouts at 1, 10 and 100 supports, respectively, rather than the three-draw main-table means. The subtype aggregate uses the corresponding ten/ten/five case design. L2 and L3 denote the supplied second and third annotation levels; sibling tasks restrict classification to a related group. These eight tasks were defined after the initial prior-screen readouts and are exploratory; they are not a formally defined "easy versus difficult" partition. The broad/coarse and fine primary panels remain in Table 1. The available checkpoint summaries do not provide CellPFN support-draw uncertainty, so no interval is inferred from the proxy-model case records. Different label sets and class counts prevent treating cross-task accuracy differences as isolated changes in difficulty.

Original ZEBRA sex-classification case.

Scoring unit	Balanced accuracy (%)	Log loss (nats)
Query cells	100.0	0.409
Aggregated GSM samples	100.0	0.398

This single Sayed-study oligodendrocyte case used 200 support cells from 12 GSM samples and 50 query cells from ten disjoint GSM samples, with 21,158 processed genes. GSM separation does not guarantee donor separation. One support draw was evaluated; the perfect observed accuracy does not imply an uncertainty-free population estimate. This case is separate from the four core cell-type panels and is excluded from their averages.

Later unit-level sex evaluation: positive control and broader task.

Support units/class	XY-eligible task AUC	Broader sex task AUC
4	0.977 [0.872, 1.000]	0.507 [0.391, 0.617]
8	0.993 [0.949, 1.000]	0.544 [0.453, 0.617]
16	1.000 [1.000, 1.000]	0.538 [0.441, 0.633]
32	1.000 [1.000, 1.000]	0.539 [0.426, 0.656]

Values are mean [minimum, maximum] unit-level AUC over ten support splits at the same reference CellPFN checkpoint, using 8,192 HVGs, ten support cells per unit and twenty query cells per unit. Units are persons where resolved; Sayed age-by-sex GSM groups remain a proxy, and cross-study person overlap is unknown. The XY-eligible positive control requires measured XIST and at least four Y-linked genes; this eligibility rule does not identify genes used by the model. The broader task remains close to chance. Splits reuse persons and are not independent cohort replications, so observed ranges are shown without confidence intervals. Support budgets of 16 and 32 units per class exceed the training support range. These later tasks are distinct from the original cell-level case above.

Sex-specific gene attribution remains unresolved. XIST and Y-linked genes define eligibility; they are not measured attribution hits.[O06]